Development of New 1.0 L I3 Gasoline Direct Injection Turbocharged Downsizing Engine

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ABSTRACT

A new 1.0 L inline 3-cylinder VTEC TURBO engine that achieves enhanced fuel economy and driving performance has been developed as part of the cluster of new-generation power train technologies known as EARTH DREAMS TECHNOLOGY, which realizes both environmental and driving performance.

This engine has the same combustion concept as the previously released 1.5 L and 2.0 L 4-cylinder VTEC TURBO engines, and also applies common technologies such as high-tumble-flow intake ports, multi-hole injectors, and valve timing control of both the intake and exhaust camshafts. In addition, technologies such as a variable displacement oil pump, timing belt in oil system, and narrow crankshaft have been applied to comprehensively reduce friction. A compact class vehicle with this engine installed achieved 26% better fuel economy in European driving mode compared to the previous engine, and a newly developed compact turbocharger helped improve driving performance with 20% higher low- and medium-speed torque. Furthermore, both vibration and noise were kept in competitive level.

1. Introduction

New and stricter target values for fuel economy regulations have been set for compliance in developed countries in Europe, North America, and elsewhere from 2020 to 2025. Developing countries are also moving to implement those targets, so OEM manufacturers in Europe, in particular, are developing turbocharged downsizing engines as a technology that both complies with those fuel economy regulations and achieves product competitiveness.

Honda has released 1.5 L[(1), (2)] and 2.0 L[(2), (3)] 4-cylinder VTEC TURBO engines as part of the cluster of new-generation power train technologies known as EARTH DREAMS TECHNOLOGY, which increases both environmental and driving performance. These VTEC TURBO engines achieve equivalent dynamic performance with a smaller displacement than conventional naturally aspirated (NA) engines, and friction has also been reduced to enhance fuel economy.

These VTEC TURBO engines are positioned as world strategy engines intended to help meet fuel economy regulations and increase product competitiveness in the European, North American, Chinese, and other markets. The newly developed 1.0 L 3-cylinder VTEC TURBO engine is also being marketed as part of this engine series. This paper introduces the construction, main technologies, and performance achieved by the developed engine.

2. Development Goals

Turbocharged downsizing engines have smaller displacements compared to an NA engine of equivalent power, so the friction is less, and use of the high-load range at low engine speeds makes it possible to reduce the pumping loss and enhance fuel economy. In addition, turbocharging provides sufficient driving performance even with smaller displacement when high power is needed during acceleration and high-speed driving. A 1.0 L 3-cylinder engine was newly developed that uses downsizing and various friction-reducing technologies to enhance fuel economy. The development goals were as follows.

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(1) 20% or greater enhancement of same vehicle class fuel economy compared to the previous 1.8 L NA engine
(2) 20% enhancement of low- and medium-speed torque
(3) Reduction of 3-cylinder characteristic vibration produced by reciprocating mass

The displacement was selected in consideration of the friction reduction rate relative to the 1.8 L NA engine, which meets the above-noted fuel economy target, and the acceleration response during starting when turbocharging does not function, which is an issue for downsizing. Figure 1 shows an overview of the developed engine.

Fig. 1 Overview of developed engine

3. Main Engine Specifications

Table 1 compares the specifications of the developed engine with those of the current 1.5 and 1.8 L NA engines. In order to maximize the utilization of existing production facilities, the main dimensions, such as bore diameter, are the same as those of the current 1.5 L NA engine. Approximately 50% of the parts were newly developed for this engine.

The main feature of the developed engine is the use of friction-reducing technologies such as a narrow crankshaft, timing belt in oil system, variable displacement oil pump, and electronically controlled thermostat. In addition, the heat mass of the exhaust system upstream to the catalyst is reduced by mounting the turbocharger directly to the cylinder head integrated with the exhaust manifold. This promotes early catalyst warm-up, reduces emissions, and suppresses the drop in fuel economy performance due to the air-fuel ratio enrichment needed during the warm-up process.

Table 1 Comparison of engine specifications

Engine specifications	Developed 1.0 L I3 turbocharged	Previous 1.5 L I4 NA	Previous 1.8 L I4 NA
Displacement [cm^3]	988	1496	1798
Bore [mm]	73	73	81
Stroke [mm]	78.7	89.4	87.3
Stroke / Bore [-]	1.08	1.22	1.08
Compression ratio [-]	10.0	11.5	10.6
Valve number In/Ex [-]	2/2	2/2	2/2
Valve diameter In/Ex [mm]	28/23	29/25	33/26
Crankshaft main/pin journal diameter [mm]	38/35	46/40	55/45
Piston with oil gallery	Applied	Not applied	Not applied
VTC	Dual VTC	In-VTC	None
VTEC	In	In	In
Variable displacement oil pump	Applied	Not applied	Not applied
Electrically controlled thermostat	Applied	Not applied	Not applied
External EGR system	None	Hot	Hot
Fuel injection system	Direct	Direct	Port
Camshaft drive	Timing belt	Timing chain	Timing chain
Fuel	RON 95	RON 95	RON 95
Power [kW/rpm]	95/5500	97/6600	106/6500
Torque [Nm/rpm]	200/2250	155/4600	174/4300

4. Construction and Achieved Technology

4.1. Cylinder Block and Cylinder Head

The cylinder block is made of die-cast aluminum alloy with an iron liner cast in. A dual cooling system structure that uses a sub-jacket formed in front of the cylinder block to split the coolant flows into two streams was used to enhance the cooling performance to compensate for the increased power density. One stream flows directly to the cylinder head, with the other flowing to the cylinder head via the cylinder block water jacket (Fig. 2).

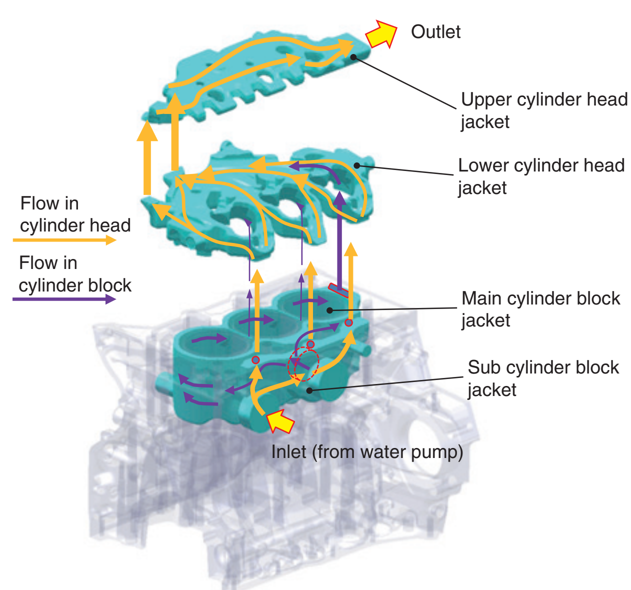


Fig. 2 Coolant flow in engine

The cylinder head uses a vertically split two-piece cooling jacket construction, and the coolant traverses the combustion chamber, enabling to evenly cool the combustion chamber of each cylinder (Fig. 3).

4.2. Cooling System

The electronically controlled thermostat shown in Fig. 4 was installed at the coolant channel outlet of the engine to both enhance fuel economy by reducing friction and to secure strength. The wax-element opening temperature of the electronically controlled thermostat is set to 103°C, which is 20°C higher than that of the conventional system. Increasing the coolant temperature under low-load conditions reduced friction by maintaining a high oil temperature, and enhanced mode fuel economy by approximately 0.6%. In addition, the wax-element piston has a built-in ceramic heater that enables opening at a lower coolant temperature than the set temperature by directly heating the wax. Under high-load conditions, power is supplied to the ceramic heater to lower the opening temperature and control the coolant temperature like in the conventional system, in order to secure strength.

4.3. Reciprocating Parts

The crankshaft uses high-strength nitrided steel, and the main journal and crankpin journal diameters were reduced in order to reduce friction. Figure 5 shows an overview of the crankshaft and compares the dimensions with those of the 1.5 L NA engine.

In addition, even vibration rooted in the first-order coupled inertia moment is a phenomenon that is a challenge to avoid in a 3-cylinder engine, a balancer-less system was used to avoid increases in friction and weight.

Countermeasures for vibration produced by the first-order coupled inertia moment included reducing the weight of reciprocating parts and setting a crankshaft balance ratio that reduces vertical vibration at the engine mount points, which greatly contributes to body vibration. Balance ratios of 0% and 75% were compared when developing this engine, and 75% was selected.

Figure 6 shows the results of comparing the vibration at powertrain mount points due to the first-order coupled inertia moment with that of competing vehicles. This

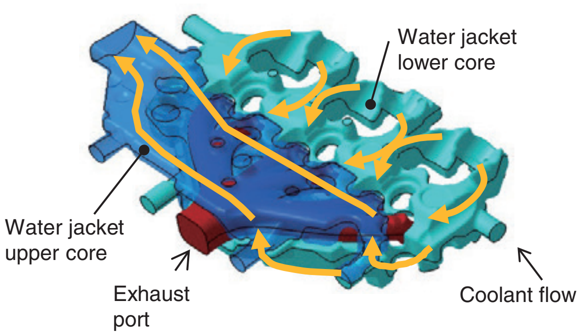


Fig. 3 Configuration of two-piece water jacket and coolant flow in cylinder head

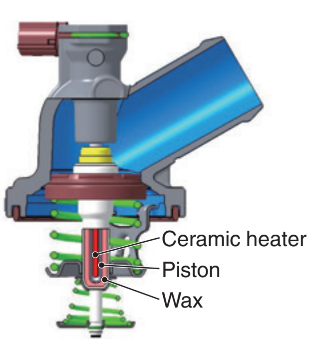
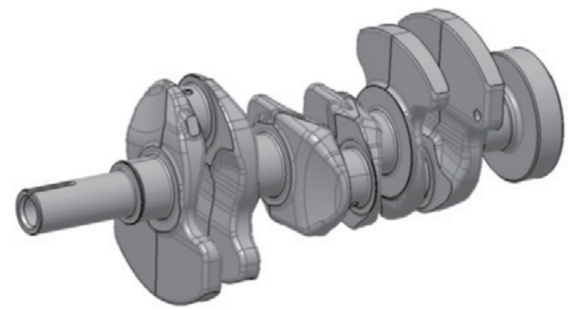


Fig. 4 Sectional view of electrically controlled thermostat



	1.0 L I3 turbocharged	1.5 L I4 NA
Pin bearing diameter [mm]	35	40
Main bearing diameter [mm]	38	46
Crank radius [mm]	39.16	44.5

Fig. 5 Crankshaft overview

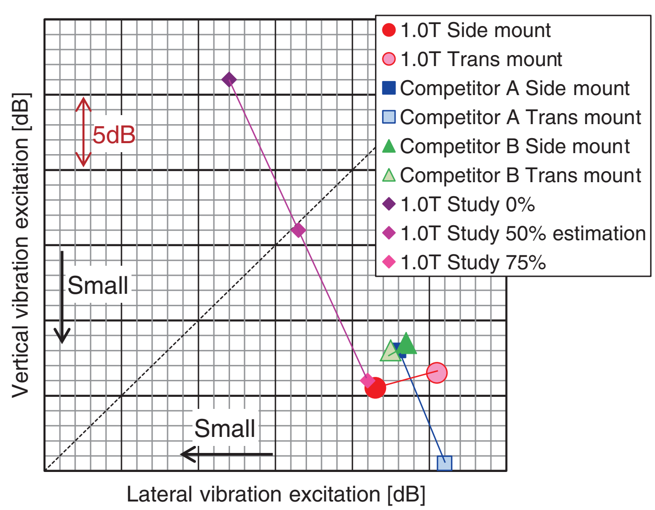


Fig. 6 1st order vibration excitation of I3 turbo charged engine at 1000 rpm confirmed that the developed engine is sufficiently competitive with rival vehicles.

The pistons use a cooling channel to compensate for the increased thermal load due to turbocharging (Fig. 7). This reduced the piston crown temperature by approximately 20°C, enhancing both strength and the knocking characteristic, leading to a higher compression ratio, thereby enhancing fuel economy by 0.5%.

Like the 1.5 and 2.0 L VTEC TURBO engines, the connecting rod used the controlled forging method to compensate for the high compression load due to turbocharging. The controlled forging method uses a two-step forging process to reinforce the I-beam area. Increasing the buckling strength of the I-beam area made it possible to reduce the weight by 18% compared to conventional forging technology.

4.4. Valve Train

The valve train uses valve timing control (VTC) for both intake and exhaust, with a VTEC rocker arm on the intake side that can switch between high and low lift. Optimal VTC and VTEC control is performed according to the operating conditions such as the load and engine speed. This enhances fuel economy under low-load conditions while still delivering torque under high-load conditions. Figure 8 shows the valve train external view.

3-cylinder engines with a short overall length have limited layout space especially when installing a turbocharger, so an efficient layout for the variable valve control system is needed. The developed engine uses a VTC structure that positions the VTC control solenoid and oil control valve (OCV) over the camshaft as shown in Fig. 9.

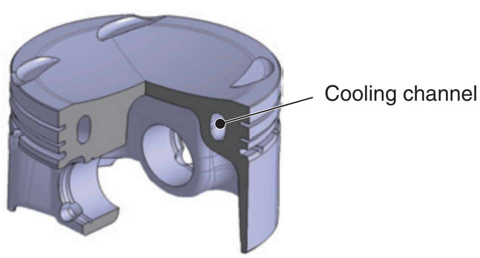


Fig. 7 Cut-off view of piston

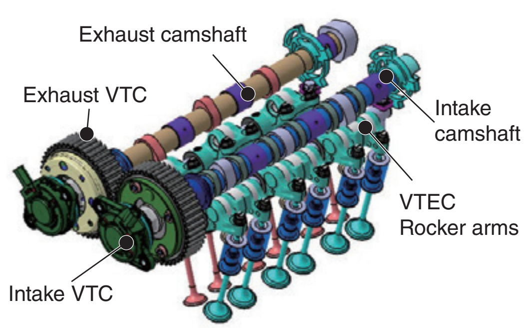


Fig. 8 Schematic view of valve train

This simplified the oil circuit, and the effects of the built-in check valve helped deliver high responsiveness relative to the VTC size.

A VTEC mechanism that switches between low and high lift is used on the intake side. In low-lift mode, fuel economy is enhanced by an early valve-closing Atkinson cycle described hereafter, which enhanced brake specific fuel consumption (BSFC) by up to 5% and mode fuel economy by 2%. Figure 10 shows the intake and exhaust valve timings and lifts, and Fig. 11 shows the VTEC and VTC switching strategy map. Regarding the application of VTEC to this engine, an early-closing Atkinson cycle valve

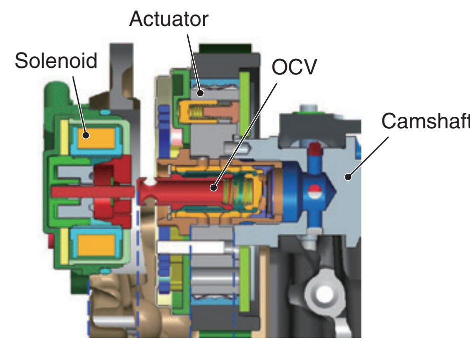


Fig. 9 Structural drawing of VTC

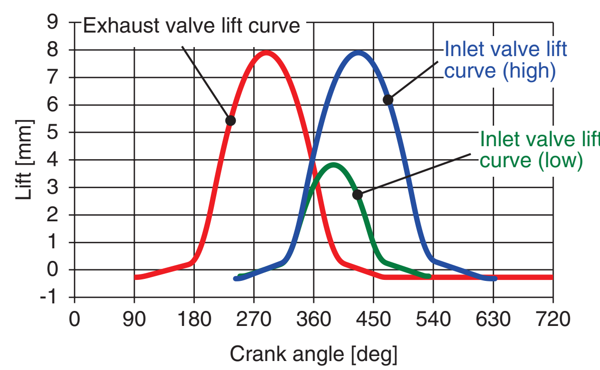


Fig. 10 Valve timing and lift

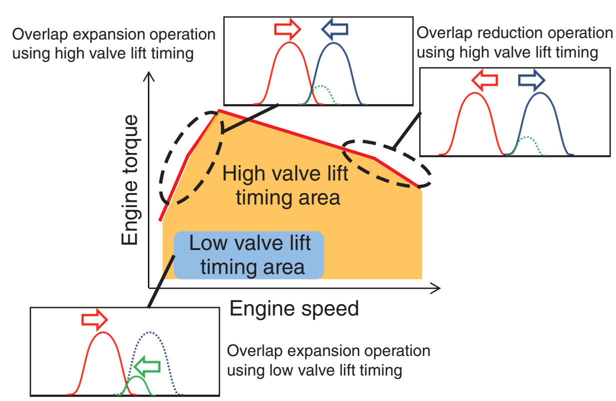


Fig. 11 VTEC and VTC switching strategy map

timing was used to enhance fuel economy under low-load conditions in consideration of the vehicle class in which the engine is mounted. While the Atkinson cycle reduces pumping loss under low-load conditions, it also restricts the valve timing and lift, which reduces the torque under high-load conditions, making it a challenge to simultaneously achieve both fuel economy and power.

Proprietary VTEC technology was used to address this issue, providing enhanced fuel economy under low-load conditions as well as sufficient torque under high-load conditions by switching the cam according to the load. Figure 12 shows the indicated diagram at each valve timing, and demonstrates that pumping loss is reduced by selecting early valve-close timing under low-load conditions.

The developed engine drives the camshaft and oil pump using timing belts located inside the engine (Fig. 13). The

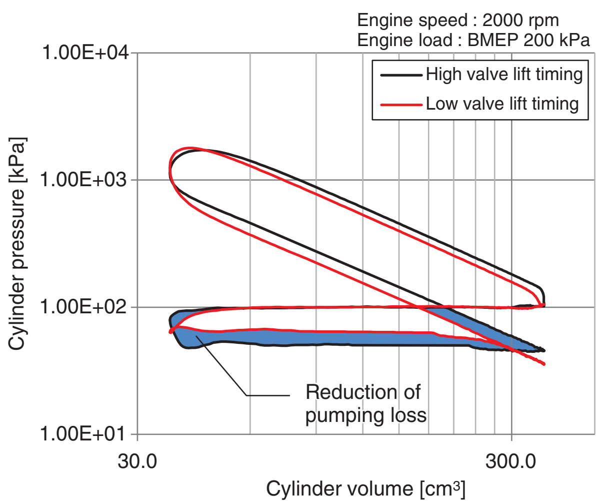


Fig. 12 Indicated diagram

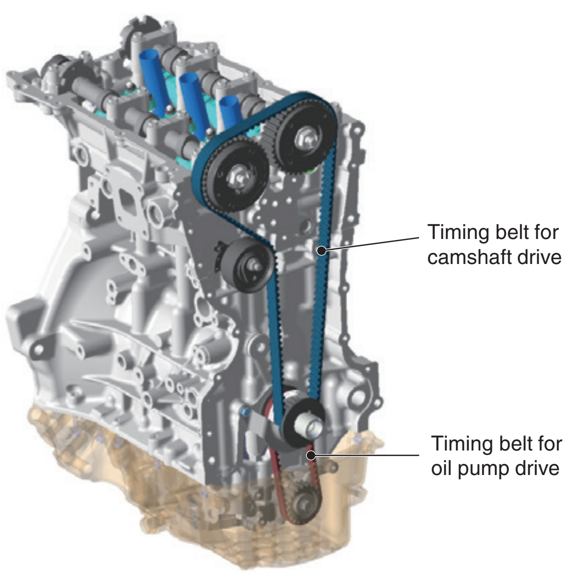


Fig. 13 Overview of timing belt system

Table 2 Comparison of timing belt materials

		Belt in oil	Conventional Belt
Tooth fabric	Material	PA66+Aramid	
	Surface treatment	Epoxy resin	H-NBR
Rubber	Material	H-NBR +Aramid fiber	
	Back face	Back face fabric	H-NBR

timing belts are located inside the engine, so the belt noise is reduced and engine friction is also reduced by 1.8% compared to a conventional timing chain. Timing belt materials were developed with specifications that provide sufficient durability for use in the oil system environment. Figure 14 shows the timing belt construction, and Table 2 compares the material specifications with those of a conventional timing belt in air.

4.5. Lubrication System

A variable displacement oil pump was used to reduce engine friction. The pump is located in the lower interior of the engine, and is driven by the crankshaft via a timing belt. An electronically controlled solenoid is used to switch the oil pressure between two levels in accordance with the engine load and speed, thereby maintaining the target oil pressure or less. This helps prevent the oil pressure from rising excessively even during the warm-up conditions of low oil temperature and high viscosity, further reducing friction and enhancing mode fuel economy by approximately 1%.

Variable oil pump capacity is achieved by use of a structure that changes the eccentricity by swinging the cam ring forming the outer ring of the vane pump (Fig. 15). Oil pressure applied to the chambers above and below the cam ring controls this eccentricity.

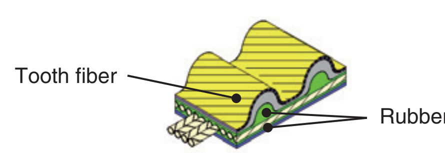


Fig. 14 Construction of timing belt

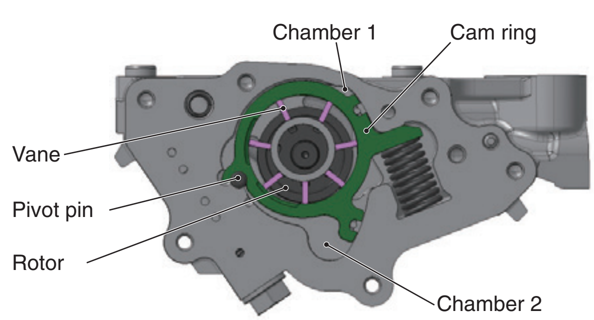


Fig. 15 Sectional view of oil pump

4.6. Turbocharger

A single scroll-type turbocharger was used (Fig. 16). Use of a compact turbine wheel reduced the inertia compared to a conventional turbocharger, and in combination with an electric waste gate actuator achieved high torque and high response from low engine speeds with enhanced fuel economy under low-load conditions. A turbine housing made of cast stainless steel offers high heat resistance, and the specifications do not rely on enriching the air-fuel ratio to reduce the exhaust temperature, even when exhaust gases are at high temperatures up to 950°C. And the higher permissible turbine speed enabled to support higher power (Table 3).

4.7. Combustion Concept

The combustion concept of the developed engine is rapid combustion like the previously released 1.5 L and 2.0 L 4-cylinder VTEC TURBO engines. This was achieved by the use of high-tumble flow ports, VTC on both the intake and exhaust camshafts, and multi-hole direct side injectors, which creates high-tumble flow with the optimal air-fuel mixture, and enhances fuel economy by reducing the time loss in the high-load range. Figure 17 compares the port and combustion chamber shapes of the developed engine and the 1.5 L NA engine. The developed engine uses a flatter intake port angle compared to the NA engine, which realizes an intake flow along the pentroof wall surface on the exhaust side (Fig. 18). In addition, by providing a small raised edge at the bottom of the intake port (Fig. 17 detail A), the main intake airflow is guided to the upper part of the combustion chamber, strengthening the tumble flow.

The direct injector hole layout and the size of each injector hole were selected to reduce fuel adherence to the piston and liner in order to suppress soot generation and dilution of the lubricating oil, and to increase the homogeneity of the air-fuel mixture to ensure rapid combustion (Table 4).

Table 3 Comparison of turbocharger specifications

Turbocharger specification		1.0 L	1.5 L
Max. power [kW/rpm]		95/5500	110/5500
Max. torque [Nm/rpm]		200/2250	203/1600-5000
Turbocharger supplier		Borg Warner	Mitsubishi
Turbine	Scroll	Mono	Mono
	Wheel diameter/vane	ϕ31/10	ϕ37/11
	Permissible speed	285000 rpm	245000 rpm
	Housing material	DIN1.4837+Nb [Austenitic cast steel]	A3K [Austenitic cast steel]
Compressor	Wheel diameter/vane	ϕ37/5+5	ϕ46/6+6
	Permissible speed	285000 rpm	241000 rpm
Waste gate	Actuator	Electric	Electric
	Valve diameter	ϕ20	ϕ28

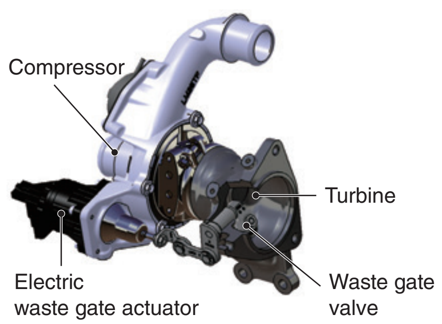


Fig. 16 Cut-off view of turbocharger

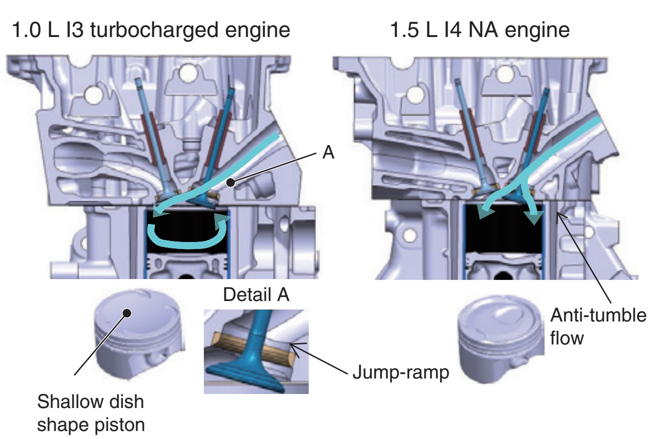


Fig. 17 Comparison of intake port and piston top shape

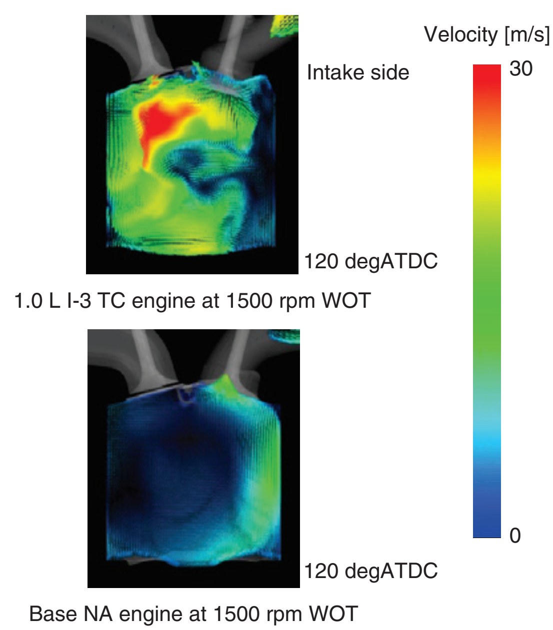
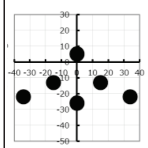
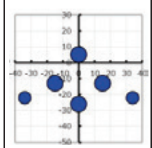
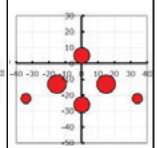


Fig. 18 Comparison of in-cylinder flow (CFD analysis)

Table 4 Injector hole configuration

	Specification at early stage	Specification at middle stage	Specification at final stage
Spray targeting and hole size arrangement			
Fuel flow distribution	Even	Uneven (small)	Uneven (large)
Piston impingement	+	++	++
Liner impingement	-	+	++
Homogeneity	+	+	+
Ranking	3	2	1

++: Good +: Average -: Poor

5. Achieved Performance

5.1. Power

This engine was developed with the goal of achieving torque characteristics equal to or better than those of the 1.8 L NA engine, to enable use in passenger car categories that include small to medium-sized vehicles. A compact turbocharger with an electric waste gate increases torque, especially in low- and medium-speed ranges. In addition, setting the optimal valve timing in each engine speed range in consideration of scavenging by VTC and internal EGR made it possible to generate a maximum torque of 200 Nm at 2250 rpm. Figure 19 compares the power characteristics with those of the 1.5 and 1.8 L NA engines. The torque at 2250 rpm is 20% higher than that of the 1.8 L NA engine, helping ensure stress-free driving. In addition, maximum power of 95 kW generated even with a displacement of only 1.0 L realizes a good balance between torque and power that can be utilized in various vehicle classes.

5.2. Fuel Economy

Figure 20 shows the position of the developed engine in terms of mechanical friction versus engine speed and mechanical friction of different displacements. Figure 21 shows the position of the developed engine in terms of BSFC at typical operating points. The mechanical friction level of the developed engine can be confirmed as near the lower limit of both engine speed and displacement trends, with BSFC positioned lowest among turbocharged engines of the same displacement[(4), (5)].

Figure 22 compares the BSFC of the developed engine with that of the 1.8 L NA engine. Optimal VTC and VTEC control achieved a BSFC of 240 g/kWh or less over a wide range, which enhanced both mode fuel economy and actual driving fuel economy by approximately 20%. Figure 23 compares the acceleration performance from 80 km/h to 120 km/h and mode fuel economy for the developed engine and the 1.8 L NA engine in the same vehicle class. This shows that mode fuel economy was enhanced by 26% while still delivering the same or better acceleration performance.

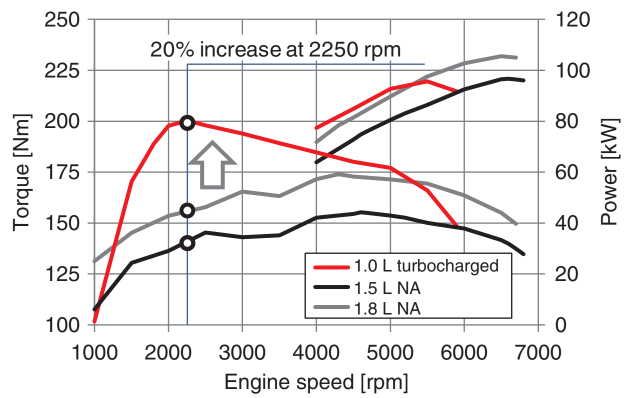


Fig. 19 Comparison of output curves

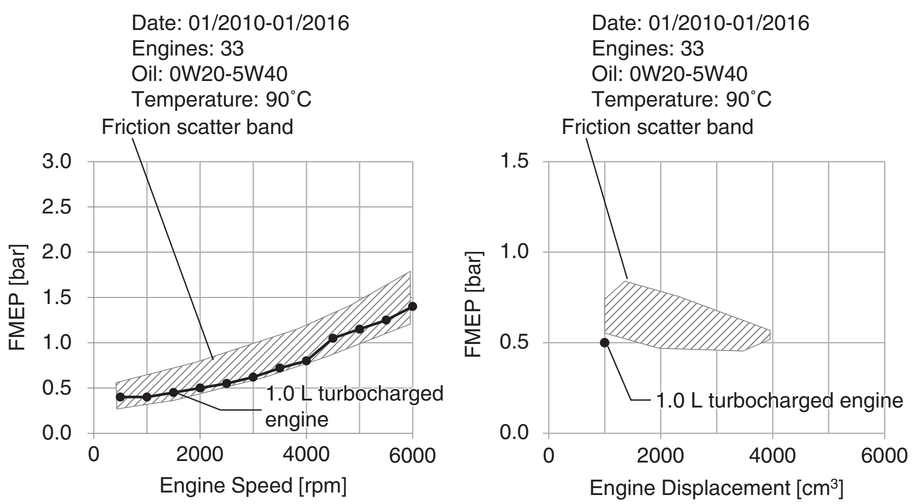


Fig. 20 Comparison of engine mechanical friction (Source of scatter band: FEV)

5.3. Emissions

In addition to hardware technology such as the abovementioned high-tumble-flow ports, the software was also modified to perform split fuel injection (up to three times) and multiple ignition (up to five times). This enhanced combustion in the cold engine condition and achieved EURO 6c level potential.

Enhanced combustion promoted catalyst warm-up by retarding the ignition timing, and decreasing the duration of inefficient catalyst warm-up also leads to enhancement of fuel economy.

5.4. NV

5.4.1. Engine radiation noise

Figure 24 shows the engine radiation noise characteristics. Turbocharged downsizing engines are known for typically increased noise at low engine speeds compared to NA engines. While the developed engine also exhibits this tendency, it is largely due to rapid combustion for the purpose of higher efficiency and increased low-speed torque.

Noise reduction measures were implemented for the main components, and also for noise radiated from the turbocharger and related piping, which does not occur in a conventional NA engine. As a result, compared to engines in the same torque and power bands, noise levels are positioned near the lower level of the turbocharged

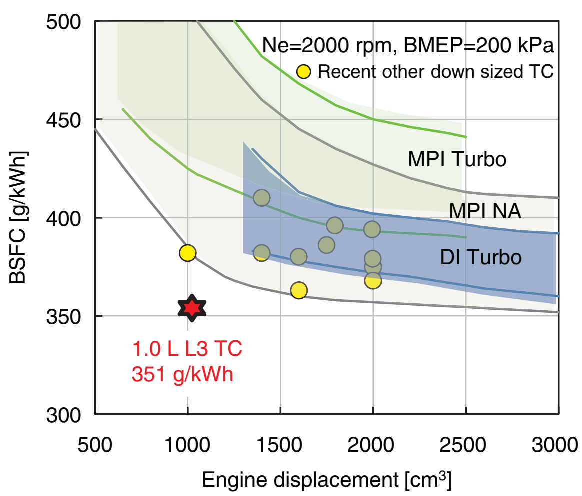


Fig. 21 BSFC (Source of scatter band: AVL)

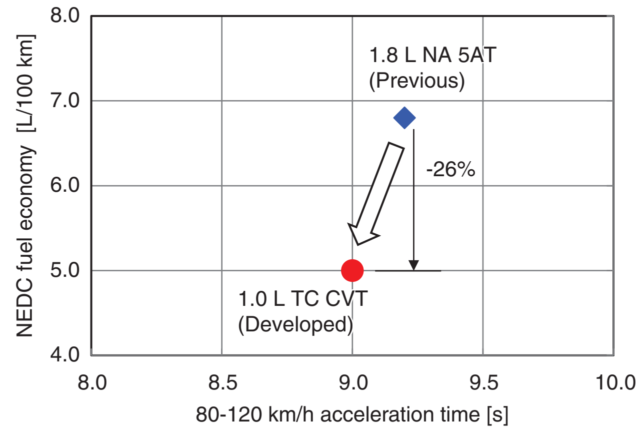


Fig. 23 Acceleration performance and fuel economy

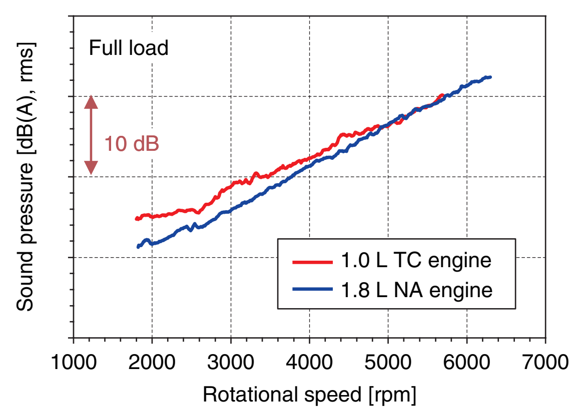


Fig. 24 Engine radiation noise

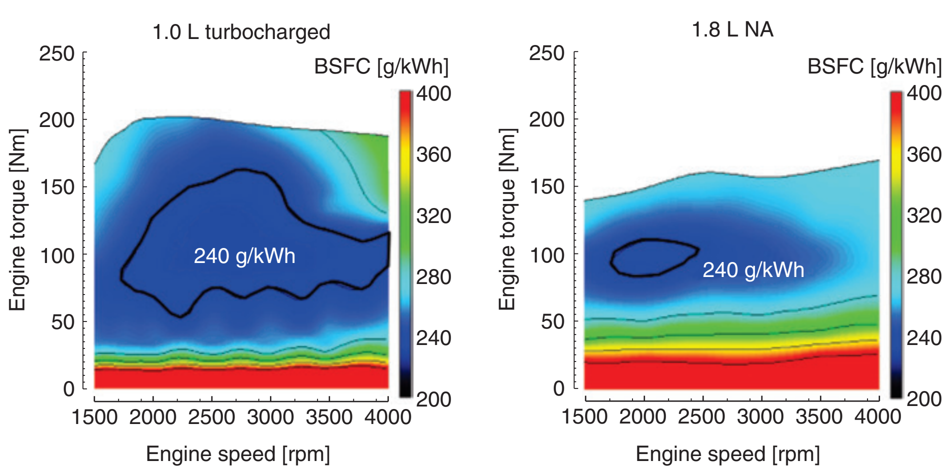


Fig. 22 Comparison of BSFC

engine group as shown in Fig. 25, demonstrating sufficient competitiveness of the developed engine.

5.4.2. Engine mount point vibration

Vibration levels are approximately in the middle compared to turbocharged engines in the same torque band, further demonstrating competitiveness (Fig. 26).

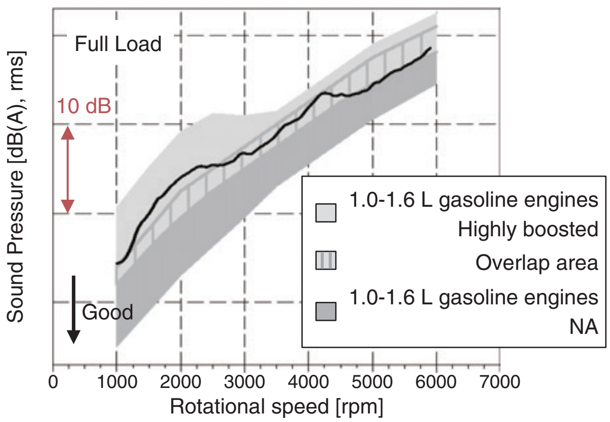


Fig. 25 Comparison of radiation noise (Source of scatter band: FEV)

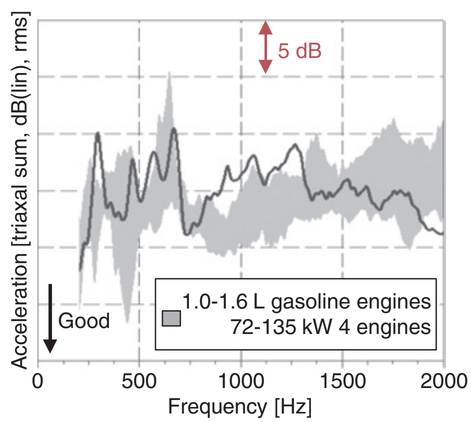


Fig. 26 Comparison of engine vibration (Source of scatter band: FEV)

6. Conclusion

The use of downsizing and friction reducing technologies obtained the following high performance potential compared to the current 1.8 L NA engine.

(1) Mode fuel economy enhanced by approximately 26% in the same vehicle class.
(2) Low- and medium-speed torque enhanced by 20% while delivering comparable power.
(3) Noise and vibration performance relative to low- and medium-speed torque help ensure sufficient product competitiveness.

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